Example 6: Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset"

Soln: p: It is sunny this afternoon q: It is colder than yesterday
r: We will go swimming s: We will take a canoe trip
t: We will be home by sunset.

Premises: 1) $\neg p \wedge q$ 2) $r \rightarrow p$ 3) $\neg r \rightarrow s$ 4) $s \rightarrow t$

Conclusion: t

Step	Reason
1) $\neg p \wedge q$	Premise
2) $\neg p$	Simplification rule on 1
3) $r \rightarrow p$	Premise
4) $\neg r \vee p$	Conditional disjunction equivalence on (3)
5) $p \vee \neg r$	Commutative law on (4)
6) $\neg r$	Disjunction syllogism on (2),(5)
7) $\neg r \rightarrow s$	Premise
8) $s \rightarrow t$	Premise
9) $\neg r \rightarrow t$	Hypothetical syllogism on (7),(8)
10) t	Modus ponens on (6),(9)

Note: Next time, try ~~only~~ to use only the rules of inference to build arguments. and generate conclusions.

Example 7: Show that the premises "If you send me an email message, then I will finish writing the program," "If you do not send me an email message, then I will go to sleep early," and "If I go to sleep